

4.3 Detecting ranges with branches (general)

PARTICIPATION
ACTIVITY

4.3.1: An if-elseif-else structure can elegantly detect ranges.

Start

☐ 2x speed

1				
2				
3				
4				
5	5 or under	No teams	If age < 6:	No teams
6				
7	Under 8	6, 7	Else If age < 8:	Play on U8 team
8				
9	Under 10	8, 9	Else If age < 10:	Play on U10 team
10				
11	Under 12	10, 11	Else If age < 12:	Play on U12 team
12	12 or over	No teams	Else:	No teams
13				

Captions ^

1. Kids of various ages may wish to play soccer. A soccer club may not have teams for kids 5
2. One level of teams is listed as "Under 8" (or just U8), which is understood to mean just 7 or
3. Likewise, U10 means 9 and 8, and U12 means 11 and 10. No teams exist for ages 12 and c
4. An if-elseif-else structure can elegantly capture such ranges. When an expression is checke
- previous expressions are False, thus defining the low-range end.

CHALLENGE
ACTIVITY

4.3.1: Decision sequence to detect increasing ranges.

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Start

Indicate the range covered by the following decision.
Assume x is a nonnegative integer.

 $x < 14$ // Range covered: - $x < 19$

Else

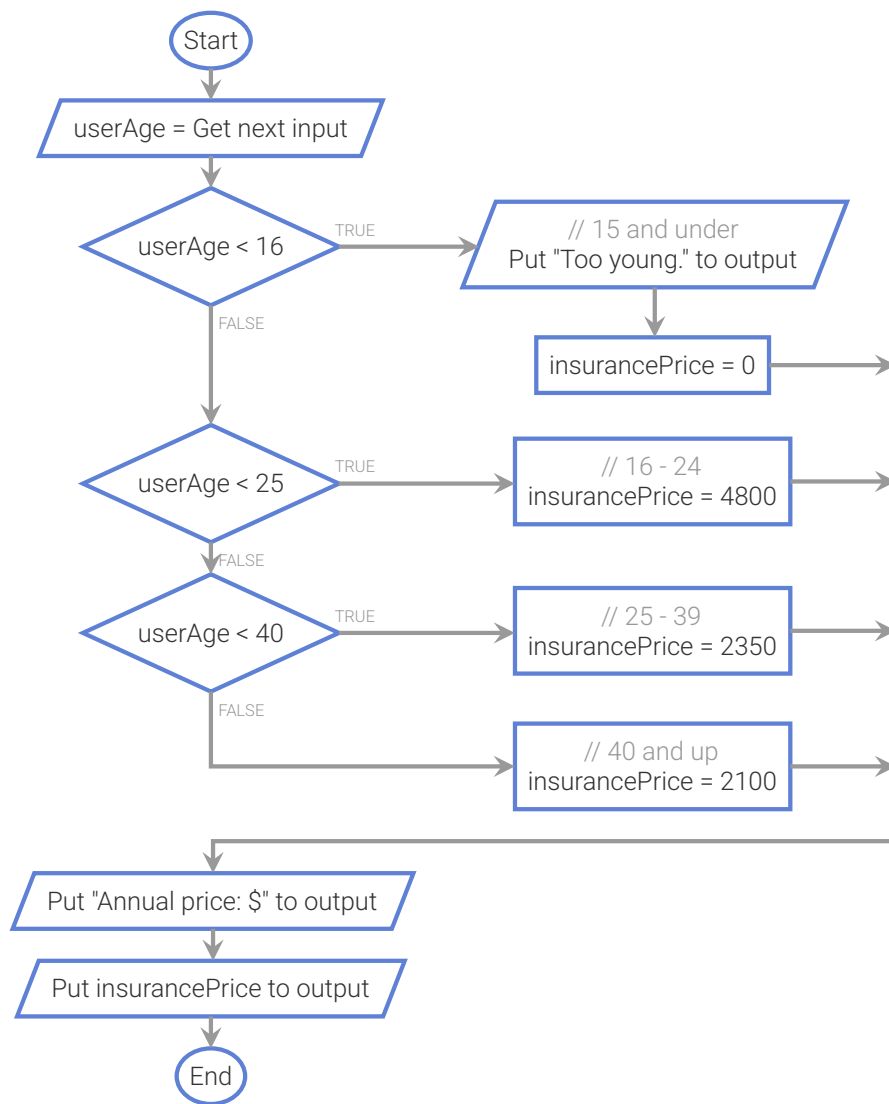
1

2

3

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ACTIVITY**

4.3.4: Using if-elseif for ranges: Insurance prices.



ENTER EXECUTION

STEP

RUN

Execution
Medium**CHALLENGE
ACTIVITY**

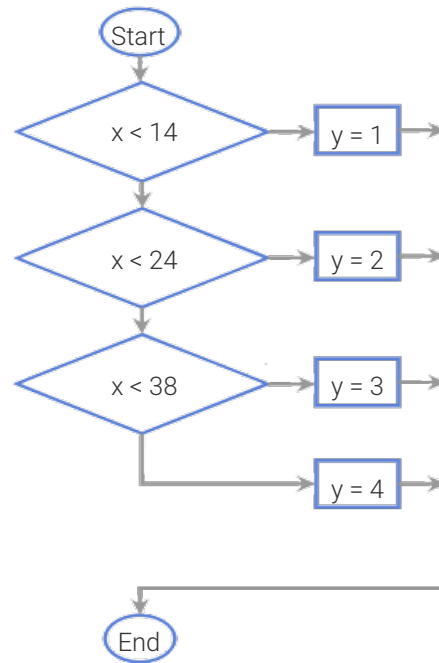
4.3.2: Flowchart decision sequence to detect increasing ranges.

763650.1722066.qx3zqy7

Start

Type the range for the given branch. Assume x is a nonnegative integer.

Range for $y = 1$: -



1

2

3

4

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